Lab5 Summary

With this lab, I gained hands-on knowledge when it comes to implementing four scheduling algorithms. I learned how each algorithm makes its own trade-offs to ensure fairness and efficiency. FCFS is relatively simple and easy to understand. The SRT algorithm focuses on shortest average turnaround time by preemptively switching to processes with less time remaining. RR ensures that there is a certain level of fairness by giving each process equal time to run, which in our case was the Quantum time of 1 second. I think the biggest thing which I noticed during lecture and in this lab as well is that RR would be one of the strongest in my opinion especially regarding the fact that it gives every process equal priority which is good in terms of ensuring fairness. The priority-based RR variantion is how I think real OS's actually combine different criterias to balance the response time of a system with the help of process priority.

Beyond just putting together the algorithms, the lab also reinforced some of my programming concepts like working with structs, managing state with static variables, and handling edge cases. I was forced to think carefully about boundary conditions like checking whether processes have arrived, whether they're finished, and ensuring remaining time is positive which would could cause minor bugs if I didn't see them. Comparing the average turnaround times across different schedulers showed me how choosing specific algorithms significantly impacts a systems performance. Overall, this lab was able to combine both theoretical concepts learnt in lectures and a practical view of how these concepts and algorithms work in real life OS's.